

Original contributions

High-resolution MR spectroscopy via intermolecular double-quantum coherences in inhomogeneous B_0 and B_1 fields[☆]

 Yanqin Lin^{a,b}, Congbo Cai^a, Shuhui Cai^a, Jianhui Zhong^{b,*}, Zhong Chen^{a,*}
^aDepartment of Electronic Science, Fujian Key Laboratory of Plasma and Magnetic Resonance, State Key Laboratory of Physical Chemistry of Solid Surfaces, Xiamen University, Xiamen, Fujian 361005, P.R. China

^bDepartment of Imaging Sciences, University of Rochester, Rochester, NY 14642, USA

Received 21 December 2010; accepted 22 February 2011

Abstract

Inhomogeneity in static field B_0 and/or RF field B_1 is inevitable under some circumstances. In this work, a method based on intermolecular double-quantum coherences is employed for high-resolution 1D MR spectroscopy via 2D acquisition under such a condition. High-resolution information on chemical shifts, multiplet patterns, J coupling constants and relative peak areas can be retained in the resulting 1D projected spectra, as shown with results from a narrow-bore NMR spectrometer and a whole-body clinical scanner.

© 2011 Elsevier Inc. All rights reserved.

Keywords: MR spectroscopy; Inhomogeneous magnetic fields; High resolution; Intermolecular double-quantum coherences

1. Introduction

Nuclear magnetic resonance (NMR) experiments are typically performed with homogeneous RF field B_1 in homogeneous static magnetic field B_0 , resulting in high-resolution MR spectroscopy with narrow resonance lines. However, there are some circumstances where B_0 and/or B_1 fields are inhomogeneous. For example, under the condition for so-called single-sided NMR, both B_0 and B_1 are inhomogeneous [1]. When surface coils are used for transmit in magnetic resonance imaging (MRI), the B_1 field varies with the distance to the coil surface [2]. In high field clinical scanners (>3 T), the homogeneity of the B_1 field, even for volume coil, is greatly degraded due to the interaction between biological tissues and electromagnetic field [3]. The homogeneity of static magnetic field B_0 is often degraded by variations in magnetic susceptibility over sample volume, such as in rock samples for oil exploration, and heterogeneous tissues in human or animal in vivo NMR

spectroscopy. Several research groups have explored various methods to address these problems. Carr-Purcell-Meiboom-Gill-like sequences, which refocus not only the effect of field inhomogeneity but also chemical shift information, were used to enhance detection sensitivity [4]. Nutation echoes that exploit a spatial correlation of the inhomogeneous B_0 and B_1 fields were employed by Pines et al. [5–7] to recover chemical shift information. Nutation echoes were also utilized by Perlo et al. [8] to develop a portable single-sided sensor for high-resolution MR spectroscopy at a low magnetic field. Antonijevic and Wimperis [9] employed two-dimensional (2D) nutation echoes.

Intermolecular multiple-quantum coherences (iMQCs) [10], which are produced by distant dipolar interactions [11,12], have caused tremendous interest [13–16]. Methods based on iMQCs have been used for obtaining high-resolution MR spectra under inhomogeneous B_0 fields. The HOMOGENIZED (HOMOGeneity ENhancement in Intermolecular ZERo-quantum Detection) sequence and similar pulse sequences based on intermolecular zero-quantum coherences were proposed [17–19]. It has been reported that there is no need to use additional phase cycling to obtain pure intermolecular double-quantum coherence (iDQC) signals, and iDQC signals are approximately 30% stronger than intermolecular zero-quantum coherence signals [20]. The IDEAL (Intermolecular Dipolar-interaction Enhanced

[☆] This work was partially supported by the NNSF of China under grants 10774125 and 10974164 and by the Fundamental Research Funds for the Central Universities (2010121010).

* Corresponding authors.

E-mail addresses: jianhui.zhong@rochester.edu (J. Zhong), chenz@xmu.edu.cn (Z. Chen).

All Lines) sequence based on iQCs was explored [21]. Recently, a sequence with great improvement in acquisition efficiency, named IDEAL-II, was developed [22]. In this study, sequences based on IDEAL-II were applied to obtain high-resolution MR spectroscopy in magnetic fields which are inhomogeneous not only in B_0 but also in B_1 .

2. Theory

iMQCs predominantly originate from spins on different molecules within a dipolar correlation distance. This distance is in the range of 5–500 μm and is usually much smaller than typical sample dimension. For a magnetic field, only the inhomogeneity within the dipolar correlation distance affects the linewidth of iMQC signal. Therefore, iMQCs can be applied to obtain high-resolution NMR spectrum in inhomogeneous static magnetic field at the scale of sample size. Moreover, the iMQC technique detects identical interactions for different spin groups in a molecule; therefore it can simultaneously retain all features of chemical shifts, multiplet patterns, J coupling constants and relative resonance peak areas.

The IDEAL-II-based sequence used in this work is shown in Fig. 1 (the pulse and gradients in brackets are used for acquisitions on a whole-body scanner only and not considered in this section). The magnetization is rotated into the transverse plane by the first pulse, and part of the solvent magnetization, which is modulated by the coherence selection gradient, is rotated subsequently back to the z -axis by the selective Gaussian pulse. This spatially modulated longitudinal magnetization gives rise to a distant dipolar field. A π refocusing pulse is inserted in the middle of the indirect evolution period t_1 like the conventional 2D J -resolved pulse sequence. One of the main purposes of this pulse is to refocus the chemical shifts of solute in the indirect (F1) dimension. Therefore, only a narrow spectral width in F1 dimension needs to be sampled, which leads to a greatly reduced data size and experimental time.

The optimal RF pulse flip angles for the IDEAL-II sequence are 90° , 90° and 180° for the first, second and last

pulses, respectively. Under this condition, when a homogeneous liquid mixture, consisting of an AX spin-1/2 system of S component with a scalar coupling constant J and a single spin-1/2 system of I component, is considered, the observable signal will be [22]

$$M^S \propto M_0^S \left[e^{-i(\Omega_1/2 - \pi J)t_1} e^{i(\Omega_S + \pi J)t_2} + e^{-i(\Omega_1/2 + \pi J)t_1} e^{i(\Omega_S - \pi J)t_2} \right], \quad (1)$$

where Ω is the frequency offset and is equal to $\omega + \gamma\Delta B$ with ω being the frequency offset in the rotating frame in the absence of field inhomogeneity, γ the gyromagnetic ratio and ΔB the spatially varying field deviation. Eq. (1) shows that the iQDC cross peaks are located at $(\gamma\Delta B/2 - \pi J, \omega_S + \gamma\Delta B + \pi J)$ and $(\gamma\Delta B/2 + \pi J, \omega_S + \gamma\Delta B - \pi J)$, respectively, when ω_1 is set to 0. This indicates that the spectral width in the F1 dimension can be as small as about half the range of B_0 field inhomogeneity, and all the peaks appear as separate streaks along the direction $\phi = \arctg(2) = 63.4^\circ$, where ϕ is the angle of spectral streaks with respect to the F1 axis. Although the ranges of the streaks in both F1 and F2 dimensions are susceptible to field inhomogeneity, an accumulated projection of the cross peaks onto the F2 dimension after rotating counterclockwise the streaks by ϕ results in a 1D high-resolution MR spectrum without inhomogeneous broadening. The relation between the iMQC signal intensity and RF pulse flip angle has been derived previously [23]. When the B_1 field is inhomogeneous, different parts of the sample will experience a different B_1 field. Therefore, not all parts of the sample will give optimal signal intensity with fixed flip angles in the pulse sequence. This will result in an overall reduced signal intensity but still retaining all high-resolution spectral features from the whole sample.

3. Experiments

Some experiments were performed on a Varian INOVA 600-MHz NMR spectrometer (Varian, Palo Alto, CA, USA) at 298 K, equipped with self-shielded x , y and z gradient coils and a 5-mm HCN triple-resonance RF coil of 1.5 cm effective length. A sample of a mixture of methyl ethyl ketone ($\text{CH}_3\text{COCH}_2\text{CH}_3$, solute) and cyclohexane (C_6H_{12} , solvent) with about 30% (by volume) acetone- d_6 was used. The liquid volume in the sample tube for the best resolution is 0.7 ml as suggested by the manufacturer. Usually, the sample tube is inserted into the spinner with its bottom below the effective range of the B_1 field (the black dashed box in Fig. 2A). To mimic a circumstance of inhomogeneous B_0 and B_1 fields, we lifted the sample tube up to place its bottom within the effective range of the B_1 field, as shown in Fig. 2A. The gradient with strength of about 0.1 T/m and duration of 1.2 ms was applied in the z direction. The selective $\pi/2$ Gaussian pulse had a width of 10 ms and was frequency selective for the solvent peak. The spectral width was 150 Hz in 75 increments for the F1 dimension and

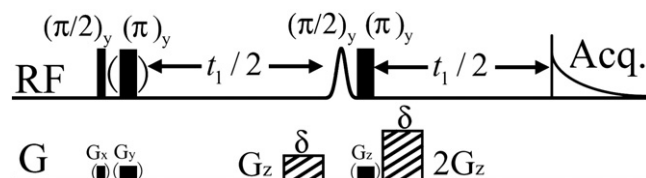


Fig. 1. IDEAL-II-based iQDC pulse sequence. Full vertical bars stand for non-frequency-selective RF pulses, Gaussian pulse is solvent selective and dashed rectangles represent correlation selection gradients. For the localized sequence used in the whole-body scanner, the pulse and gradients in parentheses were also employed. Black rectangles are slice selection gradients and hard pulses were replaced by sinc pulses. The second coherence gradient was with strength of G and duration of 2δ . The spoiler gradients are not shown.

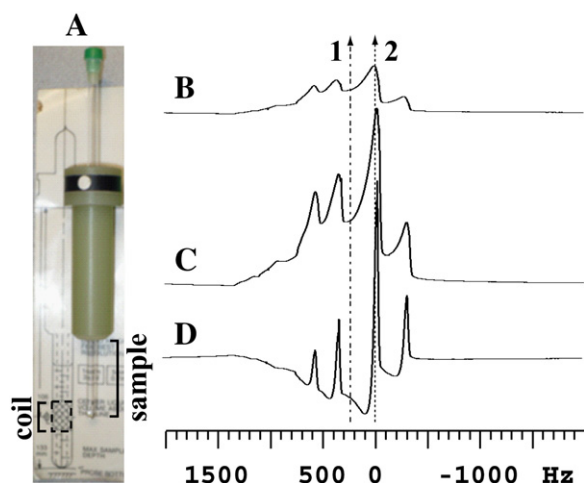


Fig. 2. (A) Sample position with coil and sample ranges indicated on a Varian vertical-bore spectrometer. The bottom of the 5-mm (outside diameter) sample tube is located in the center of the effective range of RF field (the black dashed box). 1D ^1H MR spectra of a mixture of methyl ethyl ketone and cyclohexane obtained by conventional single-pulse sequence with different pulse durations: (B) 4 μs , (C) 24 μs and (D) 44 μs . These spectra are displayed in phase mode with the same scale.

2000 Hz for the F2 dimension. The pulse repetition time was 6 s and the acquisition time was 0.6 s. To remove the unwanted coherence transfer pathways, an eight-step phase-cycling scheme was applied on the first $\pi/2$ pulse ($x, y, -x, -y, x, y, -x, -y$), the last π pulse ($x, x, x, x, -x, -x, -x, -x$) and the receiver ($x, -x, x, -x, x, -x, x, -x$). This phase-cycling scheme was also employed to select iDQCs when coherence selection gradients were void. The total experimental time was about 1 h 10 min. The acquired data were processed with the Varian spectrometer workstation software named VNMR. The iDQC signal was zero filled to 512×4096 before regular fast Fourier transform (FFT).

To test the feasibility of the technique in clinical whole-body scanners, some experiments were also carried out on a Siemens 3-T Trio whole-body scanner (Siemens Medical Solutions, Erlangen, Germany) running the 2004A software, with a standard CP birdcage head coil for RF excitation and signal reception. The gradient system is capable of producing 40 mT/m along each axis of the laboratory coordinate system with a minimal rise time of 200 μs . A proton MRS phantom in cylindrical shape with the same metabolite composition [24] and roughly the size of a normal human brain was used. It is known that the B_1 field in the central region of the head coil is relatively homogeneous, while its uniformity at the end of the coil along the z -axis is degraded [25]. In order to mimic a circumstance of inhomogeneous B_0 and B_1 fields, the phantom was placed close to or at the end of the coil and the first-order shim coil (y shimming) was intentionally misadjusted. A voxel volume of $60 \times 60 \times 60 \text{ mm}^3$ was chosen for spectral acquisition. The standard point-resolved spectroscopy (PRESS) sequence provided by the vendor was employed to acquire the conventional single-quantum signal.

The pulse repetition time (TR) and echo time (TE) were 1500 and 135 ms, respectively. The number of average was 64. The acquisition time was about 1 s with a spectral width of 1000 Hz. The total experimental time was 102 s. The IDEAL-II sequence with PRESS localization module (shown in Fig. 1) was employed to perform iDQC measurements [26]. In order to reach the steady state for magnetization, four dummy scans were acquired. The selective $\pi/2$ Gaussian pulse with a width of 8 ms was frequency selective for water. The coherence selection gradient was applied with strength of 20 mT/m and duration of 3 ms. The spectral width was 250 Hz with 128 increments for the F1 dimension and was 1000 Hz for the F2 dimension. The pulse repetition time was 1.5 s and the acquisition time was about 1 s. The total experimental time was 198 s. The raw data acquired with both localized 2D iDQC and PRESS sequences were saved and processed using custom-made software based on Matlab 6.5.0 (The Mathworks Inc., Natick, MA, USA). A sine bell shape low-pass filter was used for water suppression [27]. The iDQC signal and the PRESS signal were zero filled to 1024×4096 and 4096, respectively, before regular FFT.

4. Results and discussion

The experimental results from the Varian NMR spectrometer are shown in Figs. 2 and 3. The 1D spectra obtained using conventional single pulse sequence with different pulse duration are displayed in Fig. 2B and C. It can be seen that the spectral linewidths are broadened due to the inhomogeneous B_0 field, and little spectral information, such as chemical shifts and J coupling constants, is available. Comparing Fig. 2D to B, it can be seen that the part of the spectrum labeled as '1' in Fig. 2D experiences a RF pulse of more than 180° , while the part labeled as '2' is smaller than 180° . It is clear that the sample was in an environment with both inhomogeneous B_0 and B_1 fields when it was placed inside the spectrometer as indicated in Fig. 2A.

A 2D MR spectrum acquired with the IDEAL-II sequence in the same field as in Fig. 2 is shown in Fig. 3A (the lower part). The spectrum was then rotated counterclockwise by 63.4° (the middle part) and projected onto the F2 dimension (the upper part). It can be seen that the projection spectrum has high resolution and the information on chemical shifts, multiplet patterns, relative peak areas and J coupling constants can be obtained. This indicates the ability of the iDQC technique for high-resolution MRS in inhomogeneous B_0 and B_1 fields. It is noted that the intrinsic J coupling constant for methyl ethyl ketone is 7.3 Hz, which is scaled by a factor of 3 apparently in Fig. 3 [22].

Composite RF pulses can be utilized to partly overcome the B_1 inhomogeneity. Here the IDEAL-II sequence was performed with the first and third pulses replaced by composite pulses, e.g., $90^\circ(x)$ was replaced by $90^\circ(x) 90^\circ(y)$ and $180^\circ(x)$ by $90^\circ(x) 180^\circ(y) 90^\circ(x)$. The result in

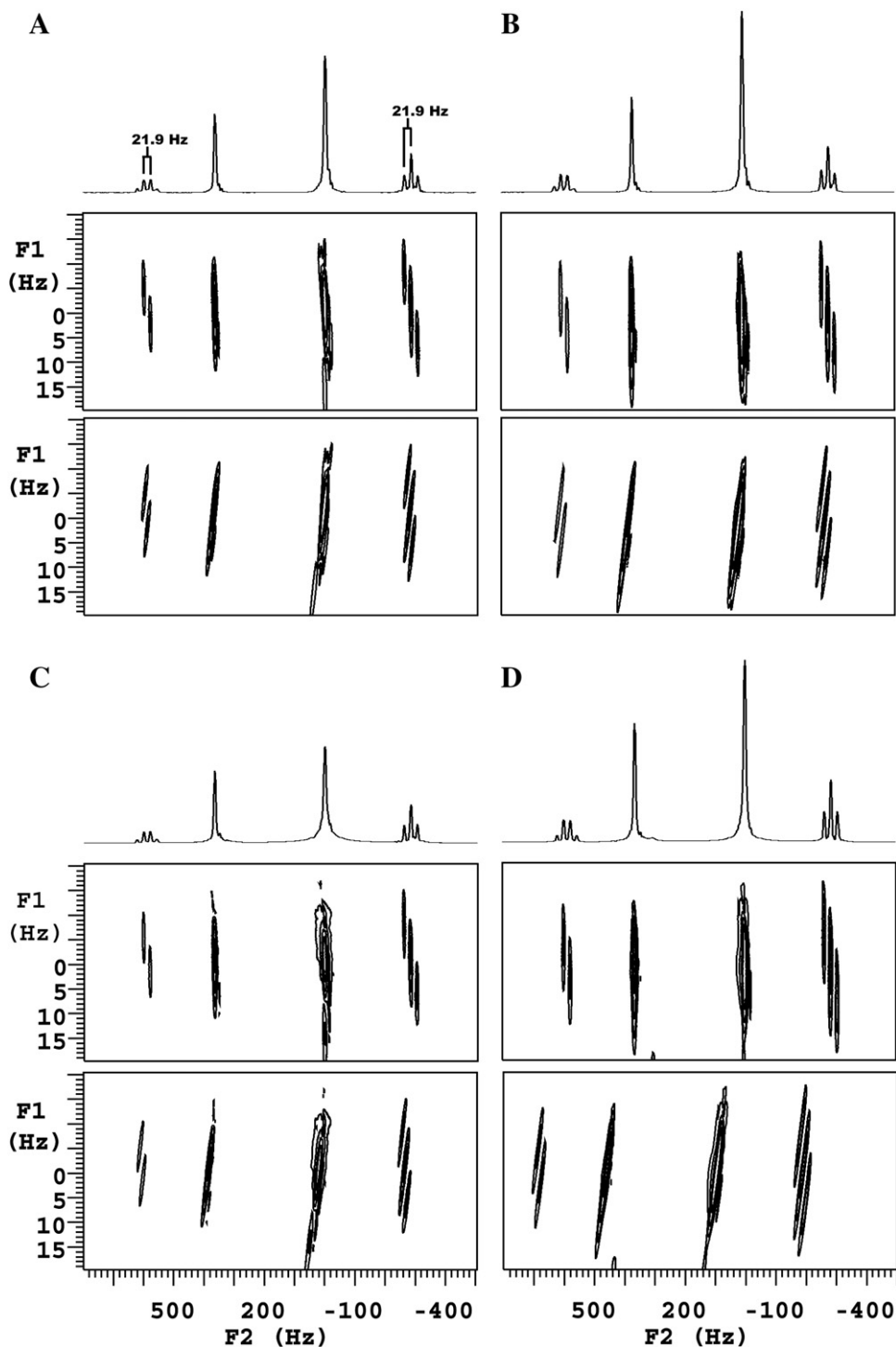


Fig. 3. Two-dimensional ^1H MR spectra of a mixture of methyl ethyl ketone and cyclohexane obtained in the same fields as for Fig. 2, their sheared spectra after a 63.4° counterclockwise rotation and corresponding 1D projections onto the F2 dimension (lower, middle and upper part, respectively). (A) With the use of the IDEAL-II sequence shown in Fig. 1, (B) is the same as (A) except that the first and last RF pulses were replaced by composite pulses, (C) is the same as (A) but without coherence selection gradients, (D) is the same as (C) except that the first and last RF pulses were replaced by composite pulses. All the spectra are displayed in magnitude mode with the same projection scale.

Fig. 3B was acquired under the same experimental conditions and processed using the same parameters as for Fig. 3A except for the use of composite pulses. It can be seen that the signal intensities in Fig. 3B are stronger than those in

Fig. 3A. It is known that iMQC signal is intrinsically much weaker than conventional single-quantum coherence (SQC) signal. Enhancement of iMQC signal intensity is crucial for practical application.

Coherence selection gradients have been considered as indispensable for high-resolution MR spectroscopy in inhomogeneous fields via iMQCs. However, it has been shown that high-resolution MR spectra can be obtained by proper phase cycling in the absence of coherence selection gradients [28]. It means that this type of experiments can be performed in an NMR equipment without gradient coils, which greatly simplifies the scanner design and leads to reduced expense. This can be very useful for potential ex situ applications. The results obtained using the IDEAL-II sequence without coherence selection gradients in the same fields as Fig. 3A are shown in Fig. 3C and D. Obviously, they have a similar resolution to Fig. 3A and the acquisition with composite pulses holds a stronger signal intensity as seen in the result in Fig. 3D compared to that in Fig. 3C.

The B_0 field inhomogeneity studied by Pines et al. [5–7,9] is in the order of thousands of Hertz. The prerequisite for their methods for high-resolution spectra is the correlation of inhomogeneous B_0 and B_1 fields, and only the information on chemical shifts is retained in the resulting high-resolution 1D and/or 2D spectra. For the iDQC method used in this study, the acceptable B_0 field inhomogeneity is tens to hundreds of Hertz. In principle, the ability of iMQCs for high-resolution spectra does not rely on the correlation of B_0 and B_1 fields. The linewidth of projected iMQC spectra mainly depends on the inhomogeneous field within the dipolar correlation distance, not across the whole sample, and not only chemical shifts, but also multiplet patterns, relative peak areas and J coupling constants can be retained in the resulting iDQC (or iMQC) spectrum. Another characteristic of iMQCs is that their signal intensity is only several to tens of percent of conventional SQC signal. Improvement of iMQC protocol was performed to enhance iMQC intensity [29]. A recent study employed nuclear hyperpolarization to yield nearly 10^6 -fold enhancement of iDQC signal [30]. This will help improve signal-to-noise ratio in the iDQC study.

The experimental results obtained from the whole-body scanner are shown in Figs. 4 and 5. The 1D PRESS spectrum in Fig. 4A was acquired at the optimal position inside the head coil with the center of voxel coinciding with that of the head coil along the z -axis for homogeneous B_0 and B_1 fields. The PRESS spectrum in Fig. 4C was obtained in inhomogeneous B_0 and B_1 fields, when the center of the voxel was 12 cm away from the center of the head coil and the static magnetic field was deshimmmed to produce a linewidth of about 30 Hz for water signal as indicated by the reading on the console during the shimming process. The corresponding sagittal anatomical image is displayed in Fig. 4B. It shows that the B_1 field is inhomogeneous across the voxel. A 2D iDQC acquisition was performed in the same inhomogeneous B_0 and B_1 fields as for Fig. 4C. The iDQC spectrum was rotated counterclockwise by 63.4° and projected onto the F2 dimension. The result is shown in Fig. 4D. It can be seen that little useful information can be obtained from the spectrum in Fig. 4C, whereas that in

Fig. 4D holds a similar resolution to that in Fig. 4A, which permits further quantification. It suggests that the iDQC method works well in such inhomogeneous B_0 and B_1 fields.

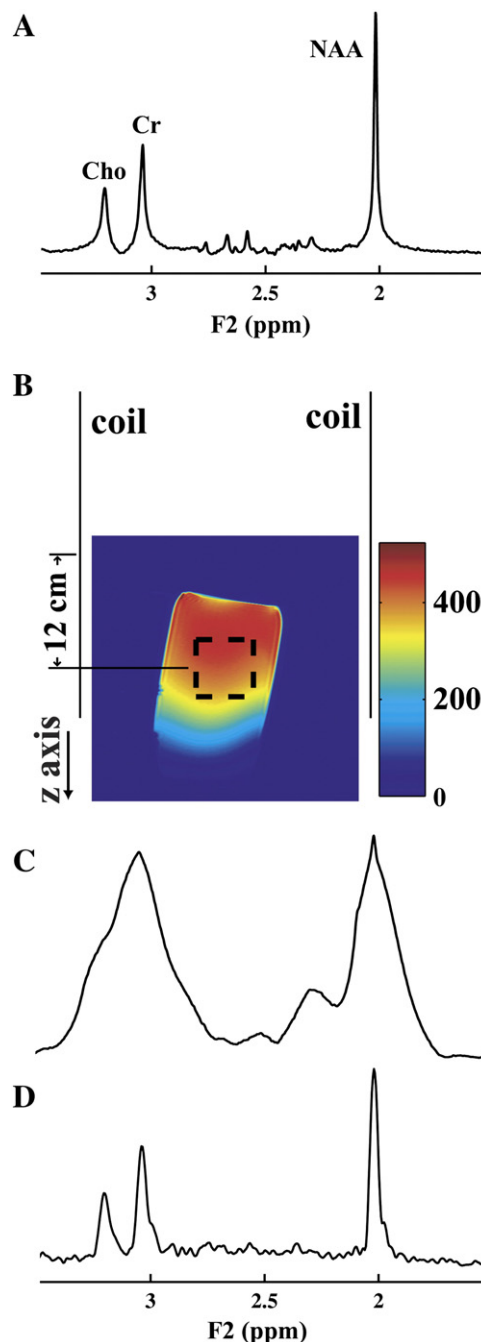


Fig. 4. One-dimensional spectra of the brain MRS phantom (A, C and D) and a sagittal anatomical image of the phantom with the center of the voxel 12 cm away from the center of the head coil along the z -axis (B) on a 3-T Siemens whole-body scanner. (A) PRESS spectrum in homogeneous B_0 and B_1 fields, with the center of voxel coinciding with that of the head coil along the z -axis, (C) PRESS spectrum obtained in the voxel in inhomogeneous B_0 and B_1 fields as indicated in (B), (D) accumulation projection from 2D iDQC spectrum acquired in the same fields as (C). In (B), the field of view is $280 \times 280 \text{ mm}^2$, the z direction is indicated by an arrow and the voxel position by a dashed box ($60 \times 60 \text{ mm}^2$). NAA=*N*-Acetyl aspartate, Cr=creatine and Cho=choline.

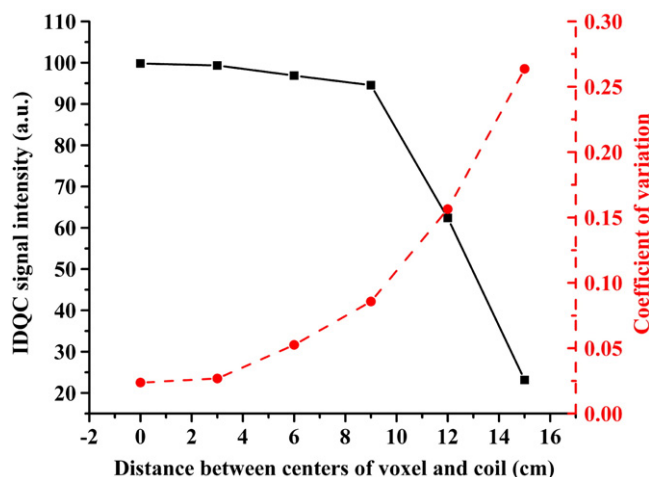


Fig. 5. Measurements on the 3-T Siemens whole-body scanner, for variation of the iDQC signal intensity and coefficient of variation with the distance between the centers of voxel and the head coil along the z direction. The x and y coordinates were kept the same during experiments. au=arbitrary unit.

To investigate the variation of iDQC signal with the degree of B_1 field inhomogeneity, iDQC measurements were performed with the voxel center located at different positions in the head coil. Fig. 5 shows that the iDQC signal intensity and coefficient of variation vary with the distance between the centers of voxel and head coil. Here, the coefficient of variation of the water signal intensity was calculated by dividing the standard deviation by the mean value of the signal in the sagittal plane of voxel (the dashed black box in Fig. 4B). It was employed to estimate the degree of inhomogeneity of the RF B_1 field. The more inhomogeneous the B_1 field, the larger the coefficient of variation. It can be seen that the iDQC signal intensity decreases with the increase of B_1 field inhomogeneity. It is noted that a wide range of B_1 intensity exists across the sample under the inhomogeneous B_1 field and the iDQC method cannot compensate the signal loss due to non-optimal RF pulse flip angle like in conventional techniques. In recent years, the trend for higher strength of magnetic field in the clinical setting continues. Higher fields will greatly increase the iDQC signal intensity due to its quadratic dependence on the field strength, but with more deteriorated homogeneity of the B_1 field. The application of the iDQC technique can be more beneficial for scanners of higher fields.

5. Conclusion

A method based on iDQCs was tested in inhomogeneous B_0 and B_1 fields on both a small-bore vertical NMR spectrometer and a whole-body clinical scanner. The method is found to be suitable for small or moderate inhomogeneous fields. High-resolution MR spectra can be obtained with information on chemical shifts, multiplet patterns, J coupling

constants and relative peak areas. This method may find applications in ex situ NMR and high-field MRI.

References

- [1] Blümich B, Perlo J, Casanova F. Mobile single-sided NMR. *Prog Nucl Magn Reson Spectrosc* 2008;52:197–269.
- [2] Pinkerton RG, Barberi EA, Menon RS. Transceive surface coil array for magnetic resonance imaging of the human brain at 4 T. *Magn Reson Med* 2005;54:499–503.
- [3] Chen J, Feng ZM, Jin JM. Numerical simulation of SAR and B_1 -field inhomogeneity of shielded RF coils loaded with the human head. *IEEE Trans Biomed Eng* 1998;45:650–9.
- [4] Hurlimann MD, Griffin DD. Spin dynamics of Carr-Purcell-Meiboom-Gill-like sequences in grossly inhomogeneous B_0 and B_1 fields and application to NMR well logging. *J Magn Reson* 2000;143:120–35.
- [5] Meriles CA, Sakellariou D, Heise H, Moulé A, Pines A. Approach to high-resolution ex situ NMR spectroscopy. *Science* 2001;293:82–5.
- [6] Heise H, Sakellariou D, Meriles CA, Moulé A, Pines A. Two-dimensional high-resolution NMR spectra in matched B_0 and B_1 field gradients. *J Magn Reson* 2002;156:146–51.
- [7] Demas V, Sakellariou D, Meriles CA, Han S, Reimer J, Pines A. Three-dimensional phase-encoded chemical shift MRI in the presence of inhomogeneous fields. *Proc Natl Acad Sci U S A* 2004;101:8845–7.
- [8] Perlo J, Demas V, Casanova F, Meriles CA, Reimer J, Pines A, et al. High-resolution NMR spectroscopy with a portable single-sided sensor. *Science* 2005;308:1279.
- [9] Antonijevic S, Wimperis S. High-resolution NMR spectroscopy in inhomogeneous B_0 and B_1 fields by two-dimensional correlation. *Chem Phys Lett* 2003;381:634–41.
- [10] He QH, Richter W, Vathyam S, Warren WS. Intermolecular multiple-quantum coherences and cross correlations in solution nuclear magnetic resonance. *J Chem Phys* 1993;98:6779–800.
- [11] Meriles CA, Dong W. Indirect detection of nuclear magnetic resonance via geometrically induced long-range dipolar fields. *J Magn Reson* 2006;181:331–5.
- [12] Dong W, Meriles CA. Indirect detection of NMR via geometry-dependent dipolar fields, revisited. *J Magn Reson* 2007;186:330–6.
- [13] Richter W, Lee SH, Warren WS, He QH. Imaging with intermolecular multiple-quantum coherences in solution nuclear magnetic resonance. *Science* 1995;267:654–7.
- [14] Barros W, Gore JC, Gochberg DF. Simultaneous measurement of D and T_2 using the distant dipolar field. *J Magn Reson* 2006;178:166–9.
- [15] Barros W, Gochberg DF, Gore JC. Nuclear magnetic resonance signal dynamics of liquids in the presence of distant dipolar fields, revisited. *J Chem Phys* 2009;130:174506.
- [16] Lin T, Sun HJ, Chen Z, You RY, Zhong JH. Numerical simulations of motion-insensitive diffusion imaging based on the distant dipolar field effects. *Magn Reson Imaging* 2007;25:1409–16.
- [17] Vathyam S, Lee S, Warren WS. Homogeneous NMR spectra in inhomogeneous fields. *Science* 1996;272:92–6.
- [18] Chen Z, Hou T, Chen ZW, Hwang DW, Hwang LP. Selective intermolecular zero-quantum coherence in high-resolution NMR under inhomogeneous fields. *Chem Phys Lett* 2004;386:200–5.
- [19] Balla D, Faber C. Solvent suppression in liquid state NMR with selective intermolecular zero-quantum coherences. *Chem Phys Lett* 2004;393:464–9.
- [20] Zhong JH, Chen Z, Kwok E. In vivo intermolecular double-quantum imaging on a clinical 1.5 T MR scanner. *Magn Reson Med* 2000;43:335–41.
- [21] Chen Z, Chen ZW, Zhong JH. High-resolution NMR spectra in inhomogeneous fields via IDEAL (intermolecular dipolar-interaction enhanced all lines) method. *J Am Chem Soc* 2004;126:446–7.
- [22] Chen Z, Cai SH, Chen ZW, Zhong JH. Fast acquisition of high-resolution NMR spectra in inhomogeneous fields via intermolecular double quantum coherences. *J Chem Phys* 2009;130:084504.

- [23] Chen Z, Zheng SK, Zhong JH. Optimal RF flip angles for multiple spin-echoes and iMQCs of different orders with the CRAZED pulse sequence. *Chem Phys Lett* 2001;347:143–8.
- [24] Schirmer T, Auer DP. On the reliability of quantitative clinical magnetic resonance spectroscopy of the human brain. *NMR Biomed* 2000;13:28–36.
- [25] Hayes CE. Radio frequency field coil for NMR. US patent 4,692,705 United States Patent and Trademark Office; 1987.
- [26] Lin YQ, Gu TL, Chen Z, Kennedy S, Jacob M, Zhong JH. High-resolution MRS in the presence of field inhomogeneity via intermolecular double-quantum coherences on a 3-T whole-body scanner. *Magn Reson Med* 2009;63:303–11.
- [27] Marion D, Ikura M, Bax A. Improved solvent suppression in one- and two-dimensional NMR spectra by convolution of time-domain data. *J Magn Reson* 1989;84:425–30.
- [28] Cai CB, Lin YQ, Cai SH, Chen Z, Zhong JH. High-resolution NMR spectra in inhomogeneous fields utilizing the CRAZED sequence without coherence selection gradients. *J Magn Reson* 2008;193:94–101.
- [29] Barros W, Gochberg DF, Gore JC. Assessing signal enhancement in distant dipolar field-based sequences. *J Magn Reson* 2007;189:32–7.
- [30] Mishkovsky M, Eliav U, Navon G, Frydman L. Nearly 10^6 -fold enhancements in intermolecular ^1H double-quantum NMR experiments by nuclear hyperpolarization. *J Magn Reson* 2009;200:142–6.